NSMO of KSA: Stage 3 Riyadh Finals & Mawhiba IMO Pathway

ANSWER KEY — COMPLETE WORKED SOLUTIONS

NSMO of KSA: Stage 3 Riyadh Finals & Mawhiba IMO Pathway Practice Exercises

1 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+2xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=2$ (with additive core $g(x)=f(x)-1x^2$ plotted). Compute $f(7)$.

a. 63

b. 49

c. 56 ✓

d. 57

Define the auxiliary function $g(x)=f(x)-1x^2$. Substituting into the functional equation cancels the $2xy$ cross-term, so $g(x+y)=g(x)+g(y)$ on $\mathbb{Z}$. Since $g(1)=f(1)-1=1$, $g(x)=1x$ and $f(x)=1x^2+1x$. Thus $f(7)=1(7^2)+(1)(7)=56$.

2 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+4xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=3$ (with additive core $g(x)=f(x)-2x^2$ plotted). Compute $f(8)$.

a. 152

b. 136 ✓

c. 128

d. 120

Define the auxiliary function $g(x)=f(x)-2x^2$. Substituting into the functional equation cancels the $4xy$ cross-term, so $g(x+y)=g(x)+g(y)$ on $\mathbb{Z}$. Since $g(1)=f(1)-2=1$, $g(x)=1x$ and $f(x)=2x^2+1x$. Thus $f(8)=2(8^2)+(1)(8)=136$.

3 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+6xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=4$ (with additive core $g(x)=f(x)-3x^2$ plotted). Compute $f(9)$.

a. 252 ✓

b. 279

c. 243

d. 225

Define the auxiliary function $g(x)=f(x)-3x^2$. Substituting into the functional equation cancels the $6xy$ cross-term, so $g(x+y)=g(x)+g(y)$ on $\mathbb{Z}$. Since $g(1)=f(1)-3=1$, $g(x)=1x$ and $f(x)=3x^2+1x$. Thus $f(9)=3(9^2)+(1)(9)=252$.

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4 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+8xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=5$ (with additive core $g(x)=f(x)-4x^2$ plotted). Compute $f(10)$.

a. 450

b. 400

c. 370

d. 410 ✓

Define the auxiliary function $g(x)=f(x)-4x^2$. Substituting into the functional equation cancels the $8xy$ cross-term, so $g(x+y)=g(x)+g(y)$ on $\mathbb{Z}$. Since $g(1)=f(1)-4=1$, $g(x)=1x$ and $f(x)=4x^2+1x$. Thus $f(10)=4(10^2)+(1)(10)=410$.

5 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(7^{27}-1^{27})$.

a. 3

b. 6

c. 4 ✓

d. 5

Verify the LTE hypotheses: $p=3$ is an odd prime, $3\nmid 7$, $3\nmid 1$, and $3\mid(7-1=6)$. Therefore, $v_3(7^{27}-1^{27})=v_3(7-1)+v_3(27)=1+3=4$.

6 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(10^{18}-1^{18})$.

a. 3

b. 4 ✓

c. 6

d. 5

Verify the LTE hypotheses: $p=3$ is an odd prime, $3\nmid 10$, $3\nmid 1$, and $3\mid(10-1=9)$. Therefore, $v_3(10^{18}-1^{18})=v_3(10-1)+v_3(18)=2+2=4$.

7 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(19^9-1^9)$.

a. 4 ✓

b. 3

c. 6

d. 5

Verify the LTE hypotheses: $p=3$ is an odd prime, $3\nmid 19$, $3\nmid 1$, and $3\mid(19-1=18)$. Therefore, $v_3(19^9-1^9)=v_3(19-1)+v_3(9)=2+2=4$.

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8 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(14^{36} - 5^{36})$.

a. 3

b. 6

c. 5

d. 4 ✓

Verify the LTE hypotheses: $p = 3$ is an odd prime, $3 \nmid 14$, $3 \nmid 5$, and $3 \mid (14 - 5 = 9)$. Therefore, $v_3(14^{36} - 5^{36}) = v_3(14 - 5) + v_3(36) = 2 + 2 = 4$.

9 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+1b+1c}{b+c} + \frac{2b+1c+1a}{c+a} + \frac{2c+1a+1b}{a+b}$.

a. 5

b. 9

c. 6 ✓

d. $\frac{9}{2}$

Splitting each fraction gives $2\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right) + 1 + 1 + 1$. By Nesbitt's Inequality (proved by adding 1 to each fraction and applying Cauchy–Schwarz / Titu), $\sum_{\text{cyc}} \frac{a}{b+c} \geq \frac{3}{2}$ with equality at $a = b = c$. Hence the minimum is $2 \cdot \frac{3}{2} + 3 = 6$.

10 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+2b+2c}{b+c} + \frac{2b+2c+2a}{c+a} + \frac{2c+2a+2b}{a+b}$.

a. 8

b. 9 ✓

c. 12

d. $\frac{15}{2}$

Splitting each fraction gives $2\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right) + 2 + 2 + 2$. By Nesbitt's Inequality (proved by adding 1 to each fraction and applying Cauchy–Schwarz / Titu), $\sum_{\text{cyc}} \frac{a}{b+c} \geq \frac{3}{2}$ with equality at $a = b = c$. Hence the minimum is $2 \cdot \frac{3}{2} + 6 = 9$.

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11 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+3b+3c}{b+c} + \frac{2b+3c+3a}{c+a} + \frac{2c+3a+3b}{a+b}$.

a. 12 ✓

b. 11

c. 15

d. $\frac{21}{2}$

Splitting each fraction gives $2\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right) + 3 + 3 + 3$. By Nesbitt's Inequality (proved by adding 1 to each fraction and applying Cauchy–Schwarz / Titu), $\sum_{\text{cyc}} \frac{a}{b+c} \geq \frac{3}{2}$ with equality at $a = b = c$. Hence the minimum is $2 \cdot \frac{3}{2} + 9 = 12$.

12 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+4b+4c}{b+c} + \frac{2b+4c+4a}{c+a} + \frac{2c+4a+4b}{a+b}$.

a. 14

b. 18

c. $\frac{27}{2}$

d. 15 ✓

Splitting each fraction gives $2\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right) + 4 + 4 + 4$. By Nesbitt's Inequality (proved by adding 1 to each fraction and applying Cauchy–Schwarz / Titu), $\sum_{\text{cyc}} \frac{a}{b+c} \geq \frac{3}{2}$ with equality at $a = b = c$. Hence the minimum is $2 \cdot \frac{3}{2} + 12 = 15$.

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